The Quadrium

Truth and Utility as Independent Measurements: A Sidecar Extension to the Triveritas

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I. ABSTRACT

This paper proposes a single-axis extension to the Triveritas framework. The Triveritas, developed within the Veriphysics tradition, evaluates the truth of claims across three independent dimensions. This paper does not modify, redefine, or reinterpret any element of the Triveritas. Whatever the Triveritas is — however its author defines it, calibrates it, or revises it — this paper accepts as the truth evaluation engine.

The extension is one axis: Utility (U).

The Triveritas evaluates whether a claim is true. It does not evaluate whether a true claim is useful, or whether a useful claim is true. Without a dedicated axis for utility, utility has no home in the evaluation. When utility has no home, it leaks into the truth evaluation as apparent truth. The mechanism is structural: an evaluator encountering a framework with extraordinary engineering output perceives the utility as evidence of truth. The perception is not irrational — useful things often are true. But useful and true are different properties, and when they share an evaluation axis, the difference becomes invisible.

The Quadrium is the Triveritas plus Utility. Three axes for truth — whatever those three axes are, as defined by the Triveritas author. One axis for utility — measuring demonstrated functional output independent of truth status. The truth axes are the Triveritas creator's domain. The utility axis is the extension proposed here.

The proposal is motivated by a specific finding documented across thirteen papers in this archive: the Standard Model of particle physics has extraordinary utility (U≈99) and zero mathematical compilation from axioms at 19 identified points (M=0). Without a U axis, the utility leaks into perceived M, making the M-failures invisible. With a U axis, both

scores are visible simultaneously. The theory is bounded, not killed. The engineering success is celebrated. The derivation failures are identified as research targets. Neither obscures the other.

This paper is a proposal to the Triveritas author, not a unilateral modification of his framework. If the author determines that a utility axis is incompatible with the Triveritas architecture, this paper documents the boundary and the proposal is withdrawn. If the author accepts the extension, whatever form it takes under his hand supersedes this proposal.

II. THE TRIVERITAS

The Triveritas is a three-dimensional epistemological evaluation framework developed within the Veriphysics tradition. Our understanding of it, used throughout this archive, is as follows.

The framework holds that any claim must survive independent scrutiny on three dimensions before meriting assent. Those three dimensions evaluate, respectively, the logical structure of the claim, the mathematical compilation of the claim, and the empirical contact of the claim with observed reality. Each dimension catches a class of error the other two cannot catch. The dimensions are independent — a high score on one does not compensate for a low score on another.

The framework identifies characteristic failure modes defined by which dimensions pass and which fail. A framework that is logically sound and mathematically rigorous but empirically untested is one kind of failure. A framework that matches data but has broken logic is another. The failure mode taxonomy provides diagnostic power that single-axis evaluation systems lack.

This description is our working understanding. It is not the definition. The definition belongs to the Triveritas author. The framework is deep — it has been validated internally through independent proofs and externally through compatibility analysis across multiple epistemological traditions spanning millennia. The reader who wishes to understand the Triveritas should go to the source: *Veriphysics: The Treatise*. We did not write it. We use it. We do not define it.

Everything that follows in this paper treats the Triveritas as a black box for truth evaluation. Whatever it contains, however it is structured, in whatever version its author publishes — that is the truth engine. Our contribution begins where truth evaluation ends and utility evaluation is needed.

III. THE GAP

The Triveritas evaluates truth. Utility is not truth.

A framework can be true and useless. Pure mathematical theorems with no application are true and have zero utility. A proof about the distribution of prime numbers is logically valid, mathematically rigorous, and empirically anchored (the primes are where the proof says they are). It builds no bridges.

A framework can be useful and untrue — or more precisely, useful and incomplete in its truth claims. The Standard Model of particle physics predicts particle interactions with extraordinary precision, guided the discovery of the Higgs boson, and undergirds the engineering of every particle accelerator on Earth. It does not derive its 19 fundamental parameters from axioms. The derivation chain is broken at 19 identified points where measured values are inserted. The utility is real. The mathematical compilation from axioms is absent at those points. Both are true simultaneously.

The Triveritas can measure the truth status — logical validity, mathematical compilation, empirical anchoring — and correctly report the M=0 at 19 points. It has no mechanism to report that the framework with M=0 at 19 points also built the modern world. The utility is invisible within the truth evaluation. It has no axis. It has no score. It has no home.

When utility has no home, it does not disappear. It leaks.

IV. THE LEAKAGE MECHANISM

Utility leaks into truth evaluation through the perception of the evaluator.

An evaluator encounters the Standard Model. The Standard Model predicted the Higgs boson — found at the LHC in 2012. The Standard Model predicts scattering cross-sections to extraordinary precision. The Standard Model's downstream predictions match measurement across the entire domain of particle physics. Every accelerator, every detector, every collider analysis relies on Standard Model calculations. The engineering works.

The evaluator perceives this utility and experiences it as evidence of truth. Not unreasonably — a framework that predicts this accurately probably has its mathematics right. The perception inflates the apparent mathematical compilation score. The evaluator does not consciously decide "utility equals mathematical truth." The evaluator experiences the combined signal of logical tightness, empirical confirmation, and engineering success as a unified impression of "this framework is correct."

The impression is coherence. Coherence is the combined L+E+U signal experienced as a single gestalt of "verified truth." The Triveritas already identifies the danger of coherence masking truth failures — this is core to its diagnostic power. The gap is that one of the components contributing to the coherence signal — utility — is not on any axis and therefore cannot be separated from the truth dimensions it contaminates.

The contamination has a specific directionality. High utility inflates perceived mathematical compilation. Low utility deflates it. The CKS corpus documented in this archive had U=0 — no engineering applications, no working systems, no functional output. When the arithmetic script found the M-failure, the kill was immediate. Zero utility meant zero resistance to the M-check. Had CKS produced a working bridge or a functioning accelerator, the M-check would have faced institutional resistance proportional to the utility at stake.

The inverse verification law — the probability of detecting an M-failure is inversely proportional to apparent M — is driven in part by utility leaking into apparent M. Higher utility produces higher apparent M. Higher apparent M produces lower probability of M-checking. The leakage is the mechanism. The axis is the fix.

V. THE FOURTH AXIS

Utility (U) measures demonstrated functional output.

U is high when the framework produces engineering applications, predictive success within operational tolerances, deployed systems, measurable real-world results. Bridges that stand. Chips that compute. Satellites that orbit. Patients who recover. Aircraft that fly.

U is low when the framework produces no functional output regardless of its truth status. Pure mathematics. Unfalsifiable theoretical constructions. Frameworks that describe but do not predict or build.

U is independent of the Triveritas output. A claim can score maximally on all three truth axes and have U=0 — a perfect theorem with no application. A claim can score zero on the mathematical compilation axis and have U=99 — the Standard Model with its 19 insertion points and its extraordinary engineering output.

The independence is the structural requirement. If U could compensate for a truth-axis failure — if high utility could excuse absent mathematical compilation — the leakage would be formalized rather than eliminated. The fix requires that U is measured on its own axis, reported as its own score, and does not trade against any truth dimension.

High U does not make a false claim true. Low U does not make a true claim false. A framework that works but doesn't compile is a framework with high U and low M. A framework that compiles but doesn't work is a framework with low U and high M. Both are informative. Both are measurable. Both require their own axis to be visible.

VI. THE STRUCTURAL PARALLEL

The extension follows a structural pattern documented across this archive.

The real number system $\mathbb{R}$ represents values as two-slot objects — numerator and denominator. Division produces three pieces of information: quotient, divisor, and remainder. Two slots for three variables. The remainder has no home. It leaks into the decimal expansion as an infinite non-terminating tail. Every irrational number is a remainder that has no slot, expressing itself through progressively smaller powers of the base because the representation system has no place to put it directly.

VDR — the exact finite discrete representation system documented in this archive — adds the third slot. Every VDR object is a triple: value, denominator, residual. Three variables, three slots. The remainder has a home. The leakage stops.

The Triveritas evaluates claims along three dimensions. Reality has four properties relevant to evaluation: the three truth dimensions and utility. Three axes for four properties. Utility has no home. It leaks into the truth axes — specifically into apparent mathematical compilation — because that is the nearest available dimension for engineering success to contaminate.

The Quadrium adds the fourth axis. Four properties, four axes. Utility has a home. The leakage stops.

The structural pattern is identical in both cases. When a representation system has fewer slots than the phenomenon has variables, the homeless variable contaminates the nearest available slot. The fix is always the same: add the slot. The mathematics and the epistemology share the structural problem. The fix is the same fix. This is either a deep structural identity or a coincidence. The parallel is stated. The reader can evaluate which.

VII. THE DIAGNOSTIC TABLE

The frameworks evaluated across this archive, reassessed with all four axes visible.

Framework	L	M	E	U	Status
QED	95	Bimodal: high discrete, zero continuous	98	99	Bounded at continuous substrate
Standard Model	90	0 at 19 points	92	99	Bounded at 19 derivation failures
General Relativity	88	0 at every structural level	90	95	Bounded at substrate, G, Λ , matter, quantum
Newtonian Mechanics	85	70	95	95	Bounded at relativistic speeds

Framework	L	M	E	U	Status
CKS (dead)	97	0	90	0	Killed — U=0, no resistance to M-check
N=1000 for pharmaceuticals	80	60	85	90	Correct tool for passive interventions
N=1000 for skill interventions	80	60	20	0	Category error — U=0 reveals mismatch
Ballet instruction	40	0	99	99	U=99 without M — skill validation
String theory	90	85	0	0	Ivory tower — compiles, builds nothing, tests nothing

The table demonstrates what becomes visible when U has its own axis.

The Standard Model and CKS have structurally identical truth profiles — high L, high E, M=0 at foundational points. They have completely different U profiles — U=99 versus U=0. The U difference explains the institutional difference: CKS was killed in 30 minutes, the Standard Model has operated for decades. The truth profiles don't explain this. The utility profile does. Without a U axis, the explanation is invisible. With a U axis, the explanation is the U score.

The N=1000 framework applied to skill-based interventions shows U=0 — it produces no valid evaluation of the skill intervention. The same framework applied to pharmaceuticals shows U=90. The category error identified in [HOWL-CULT-1-2026] is visible in the U column: same framework, different U depending on what it is applied to. Without U, both applications look equally valid on the truth axes. With U, the mismatch is measurable.

Ballet instruction has L=40 (no formal logical framework), M=0 (no mathematical model), E=99 (every dancer who trains for 15 years can perform), and U=99 (functional outcomes are undeniable). The truth axes say ballet is epistemically weak. The utility axis says ballet works. Both are true. Neither should obscure the other.

String theory has L=90 (logically rigorous), M=85 (mathematically sophisticated), E=0 (no empirical test), and U=0 (builds nothing, predicts nothing testable). The truth axes say string theory is logically and mathematically strong. The utility axis says string theory does nothing. Both are true. Without U, string theory's high L and M scores dominate the evaluation. With U, the absence of functional output is visible as a separate measurement.

VIII. BOUNDED, NOT KILLED

The Quadrium resolves the apparent conflict in [HOWL-DISC-5-2026].

That paper applied no-hedge M-accounting to the three pillars of physics and found M=0 at every foundational point where derivation from axioms is required. The finding is correct. It is also alarming in a way that obscures its own usefulness — if the Standard Model is M=0 at 19 points, does that mean we should stop using it?

No. It means we should bound it.

The Standard Model with four scores: L=90 (impressive logical architecture), M=0 at 19 points (derivation fails here, here, here — 19 research targets), E=92 (extraordinary empirical confirmation), U=99 (built the modern world). All four scores visible. All four true. No axis contaminating any other.

The M=0 points are boundaries. Each marks where derivation stops and description begins. Each is a research target — the most precise available statement of where the theory's explanation ends. Treating them as features hides the boundaries. Treating them as M-failures reveals them. The Quadrium makes both treatments visible by separating the M-score (these are derivation failures) from the U-score (the engineering works regardless).

The theory doesn't need to die. It needs to be bounded. Bounded means: published with all four scores, M-failures identified as boundaries and research targets, utility celebrated independently of mathematical completeness. The bridges stand and the M-failures are visible. Both. Simultaneously. Without contradiction.

This is the resolution the series has been building toward. Not "the Standard Model is wrong" — it isn't. Not "the Standard Model is complete" — it isn't. "The Standard Model has specific, identifiable M-failures that are hidden by its utility, and separating truth from utility on independent axes makes both visible."

IX. FAILURE MODES EXTENDED

The Triveritas identifies failure modes by which truth dimensions fail. The Quadrium extends the taxonomy with U-dependent modes.

The utility trap. High U, low M. A framework that works but doesn't compile from axioms. The Standard Model's profile. The framework is enormously productive. The derivation is incomplete. The utility masks the incompleteness. The trap is that no one checks M because U is so high. The Quadrium makes the trap visible by showing both scores: U=99, M=0. The evaluator sees the trap rather than falling into it.

The ivory tower. High M, low U. A framework that compiles but does nothing. String theory's profile. The mathematics is rigorous. The framework makes no testable predictions and builds nothing. The compilation is real. The utility is absent. The Quadrium shows both: M=85, U=0. The evaluator sees that mathematical rigor alone does not produce functional knowledge.

The cargo cult. High U claimed, low everything else. A practice that appears to work but has no logical, mathematical, or empirical support. The Quadrium shows the profile: L low, M low, E low, U claimed-high. The evaluator can distinguish between demonstrated utility (measured U) and claimed utility (asserted U without supporting evidence).

The dead framework. All axes evaluated, M killed, U=0. The CKS profile. No resistance to the M-check because no utility is at stake. The kill is clean, fast, and complete. This is what honest falsification looks like when utility does not obstruct it.

Each failure mode is defined by the specific pattern of scores across four axes rather than three. The additional axis does not complicate the taxonomy — it resolves cases that the three-axis taxonomy cannot distinguish. The Standard Model and CKS have the same three-axis profile. They have different four-axis profiles. The fourth axis is what distinguishes them.

X. THE COHERENCE TRAP RESOLVED

The coherence trap documented in [HOWL-INFO-5-2026] and [HOWL-DISC-5-2026] operates because utility leaks into apparent truth. With U on its own axis, the mechanism is visible and therefore resistible.

The evaluator encountering the Standard Model with four visible scores sees: this framework has U=99 and M=0 at 19 points. The utility is real. The mathematical compilation is absent at those points. Both are true. The evaluator is not tricked by the utility into perceiving mathematical completeness because the two properties are on different axes with different scores. The coherence signal — the combined impression of "this framework is correct" — is decomposed into its components. L is measured. M is measured. E is measured. U is measured. The gestalt is replaced by four independent measurements. The trap requires the gestalt. The four-axis decomposition dissolves the gestalt.

The inverse verification law is mitigated. The probability of checking M is inversely proportional to apparent M. When U leaks into apparent M, high utility drives up apparent M and drives down M-checking probability. When U is on its own axis, apparent M is determined by L and E alone — without the utility signal. The apparent M is lower. The M-checking probability is higher. The evaluator is more likely to check the arithmetic because the utility is not inflating the perception that the arithmetic has already been checked.

This mitigation is not elimination. L and E can still produce high apparent M without U's contribution. The Triveritas already addresses this — it is the core diagnostic the framework was built for. The Quadrium removes one contamination source (U leaking into apparent M) and leaves the Triveritas to handle the remaining sources (L and E producing coherence) with its own tools.

XI. THE SIDECAR PRINCIPLE

This paper does not modify the Triveritas. It attaches a sidecar.

The Triveritas is a vehicle for truth evaluation. It was designed for that purpose. It works for that purpose. Its internal structure — how many dimensions, what they are called, how they are calibrated, how they interact, what failure modes they identify — is the creator's domain. This paper does not open the hood.

The sidecar is U. It rides alongside the Triveritas. It measures a property the Triveritas was not designed to measure. It does not interfere with the truth evaluation. It does not modify any truth axis. It does not change any calibration. It adds one measurement that was previously leaking into the truth measurement because it had nowhere else to go.

If the Triveritas author revises his framework — adds dimensions, recalibrates, restructures — the sidecar attaches to the revised version. Whatever the truth engine becomes, U rides alongside it. The sidecar is indifferent to the internal structure of the vehicle. It measures utility. The vehicle measures truth. Both run simultaneously. Both produce independent measurements. The measurements do not interact.

The name "Quadrium" is proposed for the combined system — the Triveritas plus U — acknowledging that the truth evaluation is the Triveritas author's contribution and the utility axis is the contribution proposed here. If the author prefers a different name, a different structure, or a different integration, his determination supersedes this proposal. The finding that truth and utility are different properties requiring different axes is offered as the contribution. The implementation is the author's call.

XII. FALSIFICATION CRITERIA

F1. If U is shown to be derivable from the Triveritas dimensions — if utility is a function of truth — then U is not independent and the fourth axis is redundant. The sidecar is unnecessary because the vehicle already carries the information.

F2. If adding U as an explicit axis does not reduce the observed contamination of truth evaluation by utility in evaluator behavior — if evaluators with four visible scores are equally susceptible to the utility trap as evaluators with three — the contamination has a different mechanism and the fourth axis does not address it.

F3. If the Triveritas author determines that a utility axis is incompatible with the Triveritas architecture for reasons internal to the framework, the extension is rejected on architectural grounds. The proposal is withdrawn. The boundary is documented.

F4. If a theory is identified that has high U and maximal truth scores simultaneously at every level — no insertion points, no measured parameters, full compilation from axioms, and full engineering deployment — the claim that U and truth scores are typically divergent is weakened. The axes remain independent but the diagnostic value of the separation is reduced.

F5. If the structural parallel to VDR is shown to be analogical rather than structural — if the "information leaking into adjacent dimensions" mechanism is not the same mechanism in both cases — the parallel is ornamental rather than foundational. The Quadrium still stands on its own argument but loses the structural parallel as supporting evidence.

Each criterion is specific, testable, and stated before the evidence is examined.

XIII. CONCLUSION

The Triveritas measures truth. It does so across three independent dimensions that its author has defined, calibrated, and validated. This paper does not modify any element of that measurement. Whatever the Triveritas is, we accept it as the truth engine.

The extension is one axis: Utility. Utility measures demonstrated functional output — engineering applications, operational deployment, measurable real-world results. Utility is not truth. Truth is not utility. Both are real properties of frameworks. Both deserve measurement. Both require their own axis to be measured without contaminating the other.

Without a U axis, utility leaks into truth evaluation. The mechanism is specific: high utility inflates perceived mathematical compilation, reducing the probability that mathematical compilation is independently checked. The Standard Model at U=99 has operated with M=0 at 19 foundational points for decades without those points being treated as failures. The CKS corpus at U=0 was killed in 30 minutes when M=0 was detected. The difference is U. The U difference is invisible within a three-axis truth evaluation. It is visible within a four-axis evaluation that includes utility.

The Quadrium — the Triveritas plus Utility — makes both truth and utility visible simultaneously. The Standard Model is L=90, M=0 at 19 points, E=92, U=99. All four scores are true. All four are visible. The M-failures are research targets. The utility is celebrated. Neither obscures the other. The theory is bounded, not killed.

The structural parallel to VDR is noted: when a representation system has fewer slots than the phenomenon has variables, the homeless variable contaminates the nearest available dimension. The fix is always the same — add the slot. Whether this parallel is deep structure or surface analogy is left for the reader and for future investigation.

The proposal is offered to the Triveritas author. If he accepts the extension, whatever form it takes under his hand supersedes this paper. If he rejects it, the boundary is documented and the proposal is withdrawn. The finding — that truth and utility are different properties requiring different axes — stands independent of the implementation. The implementation is his call.

The Triveritas is his. The sidecar is ours. The road is shared.

END HOWL-SOPH-2-2026

Registry: [HOWL-SOPH-2-2026]

Status: Proposal

Domain: Applied Philosophy / Epistemology

Central Argument: Truth and utility are independent properties requiring independent measurement axes; without a dedicated utility axis, utility contaminates truth evaluation by leaking into perceived mathematical compilation

Extension Proposed: One axis (Utility) attached as a sidecar to the Triveritas truth evaluation framework, measuring demonstrated functional output independently of truth status

Key Demonstration: The Standard Model and CKS have identical three-axis truth profiles (high L, high E, M=0) but completely different utility profiles (U=99 vs U=0); the fourth axis is what distinguishes them

Structural Parallel: Same structural fix as VDR adding a third slot for remainders — information without a home contaminates the nearest available dimension; add the slot, stop the leakage

Governing Principle: The Triveritas is the author's; whatever it becomes, we accept and use; the sidecar is ours; the implementation is his call

Conclusion: Bounded, not killed — the resolution for every framework with high utility and incomplete truth

[**HOWL-SOPH-2-2026**] The Quadrium. Github: https://github.com/ghowland/papers/tree/main/papers/SOPH/HOWL-SOPH-2-2026

[**HOWL-DISC-1-2026**] The Discovery Process. Github: https://github.com/ghowland/papers/tree/main/papers/DISC/HO
DISC-1-2026

[**HOWL-DISC-5-2026**] The Institutional Coherence Gap. Github: https://github.com/ghowland/papers/tree/main/papers/D
DISC-5-2026

[**HOWL-CULT-1-2026**] The N=1000 Fallacy. Github: https://github.com/ghowland/papers/tree/main/papers/CULT/HO
CULT-1-2026

[HOWL-INFO-5-2026] Coherence as Information Phenomenon. Github: https://github.com/ghowland/papers/tree/main/p
INFO-5-2026